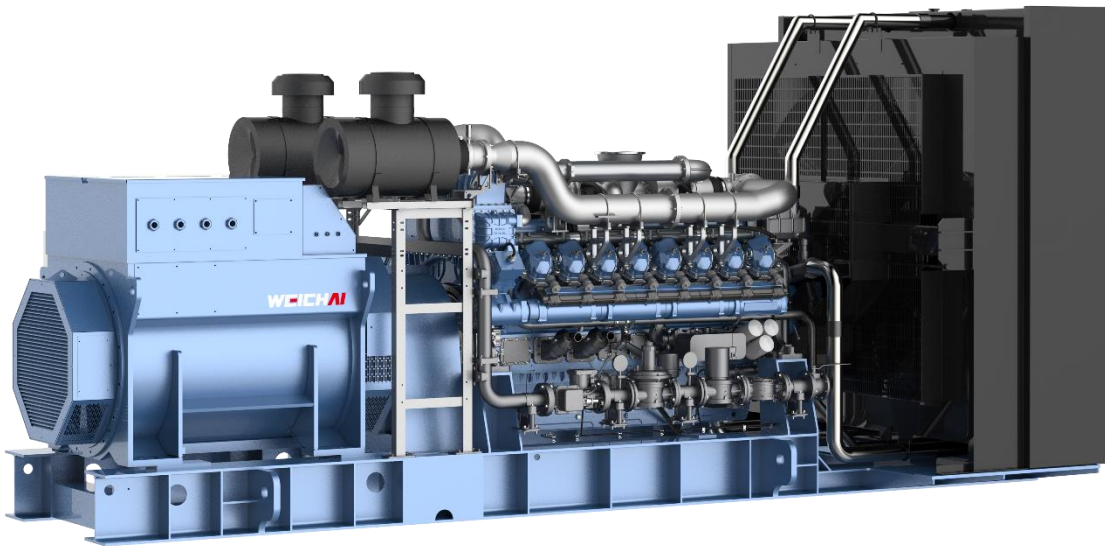


潍柴原装燃气机组数据单



型号-WPG1375*7NG

发电机组技术规格

机组型号	WPG1375B(C)7NG
额定容量/功率 kVA/kWe	1375/1100
持续容量/功率 kVA/kWe	1375/1100
额定负荷点电效率 (%)	40.2
额定电压 V	400/230
频率 Hz	50
功率因素	0.8(滞后)
相数	3
噪声@7m (dB(A))	开式: ≤96dB (A) ; 静音式: ≤85 dB (A)
环境温度 (°C)	-10~40 (若<-10°C, 增加油、水加热; 若>40°C, 功率修正)
海拔高度 (m)	≤2000 (若超出, 需进行功率修正)
燃气气源要求 (在发动机进气口 1 米范围内测量)	1、燃气温度-10~50°C; 2、燃气湿度≤80%; 3、甲烷数 MN≥85; 4、低热值不低于 32MJ/m³; 5、热值变化率小于 5%/min; 6、阀前压力变化率≤2kPa/min; 7、H₂S≤20ppm; 8、总硫(以硫计), ≤100mg/m³; 9、粉尘粒度≤5μm; 10、杂质含量≤20mg/Nm³; 11、轻质油含量≤100mg/m³; 注: 天然气体积的标准参比条件是 101.3kPa, 20°C。
燃气供气压力 (kPa)	最大入口压力为 50 (特殊压力, 请联系工厂)

机组标配

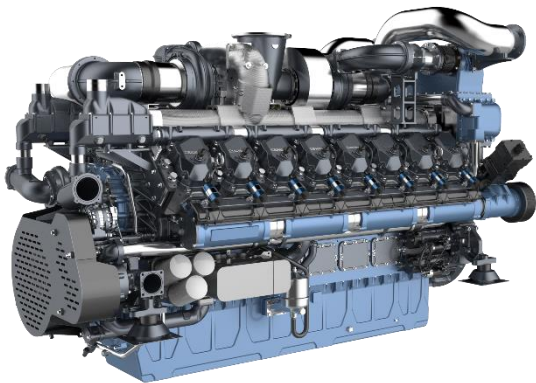
◆ 电控 ECU	◆ DC24 电启动	◆ 潍柴 WHC9510N
◆ 闭式水冷	◆ 空滤器	◆ 消音器
◆ 燃气阀组	◆ 起动蓄电池+连接线	◆ IP23
◆ 减震器	◆ PMG 励磁	◆ 发电机防冷凝加热
◆ 独立开关柜	◆ 电机轴承温度监测	◆ 电机绕组监测
◆ 颜色	开式潍柴蓝 (B) / 集装箱白色 (C)	
◆ 包装	热缩膜+篷布	

机组选配

◇ 电压 380V/400V	◇ 电压 415V/440V	◇ 高压 6.3kV/10.5kV /13.8kV
◇ 单相电压 220V/230V/240V	◇ 自切换功能(ATS)	◇ 自并联功能
◇ 水加热		

发动机

- 潍柴 16M33NG 系列，16 缸，4 冲程，闭式水冷
- 进排气系统
 - 空滤保养指示器
 - 中冷器
 - 增压器
 - 消音器
- 机油滤清器
- 燃气减压阀
- 燃气过滤器
- 40° C 散热器
- DC24V 电启动
- 蓄电池充电
- DC24V 启动蓄电池



发电机

- PMG，满足 IEC 60034-1
- 长效密封轴承
- 高效率
- 钢制外壳
- 双轴承结构
- 防护等级 IP23
- 绝缘等级 H
- 自动电压调节器



控制模块

潍柴 WHC9510N 自启动控制模块

- LCD 背光显示屏
- 发电机和市电 3 相电压检测
- 检测转速、频率、电压、电流、油压、水温
- 显示发电机组停机和报警列表
- 发动机运行时间及功率数准确监控和保养信息



发动机技术规格

发动机品牌	潍柴		
发动机型号	16M33D1280NG10		
额定转速 r/min	1500		
常用功率 kW	1280		
冷却方式	风扇闭式水冷		
调速方式	电控 ECU		
进气方式	增压中冷		
形式	16 缸、V 型		
缸径 x 行程 (mm)	150x185		
排量 (L)	52.3		
起动系统	24VDC		
冷却液容量@发动机/散热器(L)	130/NA		
散热器的设计温度 (°C)	40		
润滑油容量 (L)	171		
发动机负荷率 (%)	100	75	50
燃气消耗率 (kg/h)	230.2	178.1	126.2
发动机缸套水热量 (kJ/s)	560.0	469.2	369.9
发动机缸套水流量 (m³/h)	92.9	92.8	92.8
发动机缸套水进/出水温度 (°C)	85.1/91.3	86.3/91.5	86.8/90.9
发动机缸套水进/出水压力 (kPa)	-5.8/109	-5.9/110	-6.1/110
排气带走的热量 (kJ/s)	832.5	681.2	492.5
排气质量流量 (kg/h)	6264.0	4848.1	3339.2
涡轮后排气温度 (°C)	459	475	502
排气背压 (mbar)	75		
经余热回收后推荐排烟温度 (°C)	热水 120/0.5Mpa 蒸汽 190/1MPa 蒸汽 220		

发电机技术规格

发电机品牌	潍柴牌
发电机型号	WHA-1375-4/0.4
励磁系统	PMG
防护等级	IP23
电压调整范围	≤±1%
绝缘等级	H
绕组节距	2/3
支撑方式	双轴承

功率定义

基本功率(PRP):

在商定的运行条件下并按制造商规定的维修间隔和方法实施维护保养，发电机组每年运行时间不受限制地为可变负载持续供电的最大功率。

持续功率（COP）:

在商定的运行条件下并按制造商的规定进行维护保养，发电机组以恒定负荷持续运行并且每年运行时树不受限制的最大功率。

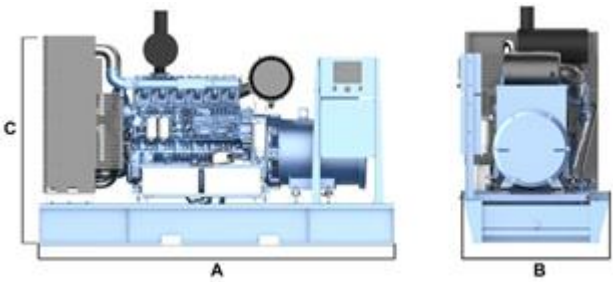
标准条件

标准机组使用环境温度要求: 环境温度-10℃~40℃，海拔高度≤1000m,相对湿度小于 90%（25℃），无沙尘、粉尘、盐雾、霉菌、凝露环境等。燃气消耗率基于燃气密度 0.726kg/N m³ 和低热值 47.3MJ/kg. 如使用环境超出以上要求时，请和工厂联系并咨询。

机组尺寸					
型号	长 A(mm)	宽 B(mm)	高 C(mm)	湿重 (kg)	备注
WPG1375B7NG	6090	2300	2900	12300	
WPG1375C7NG	12192	2438	2896	19900	

注：该尺寸不能用于安装设计，如需安装设计请根据订单咨询工厂。

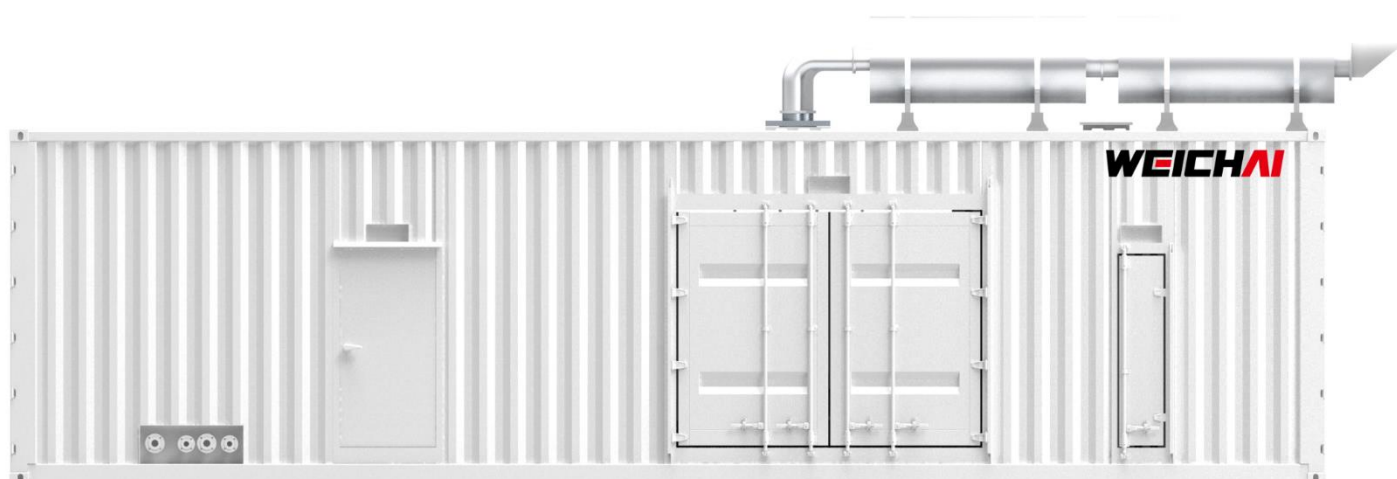
开式



集装箱式



WEICHAI GAS GEN-SETS DATASHEET



Domestic Land Based Genset– WPG1375*7NG

Gen-set specifications

Model	WPG1375B (C) 7NG
Prime Rating kVA/kWe	1375/1100
Continual Rating kVA/kWe	1375/1100
Generator set electrical efficiency at 100% (%)	40.2
Voltage V	400/230
Frequency Hz	50
Power Factor	0.8(lagging)
No. of Phases	3
Sound@7m (dB(A))	Open: ≤96dB (A) ; Silence: ≤85 dB (A)
Ambient temp (°C)	-10 ~ 40 (If < -10, oil and water heating; If > 40, power correction)
Altitude(m)	≤1500
Gas quality requirements (measure within 1 meter of engine gas inlet)	1.Gas inlet temperature:-10 ~ 50°C; 2. Gas humidity:≤80%; 3.MN≥85; 4.Pipeline natural gas with a heating value of 32MJ/m³ (or above.); 5.Rated of Heating Value;6.Pressure change rate ≤2kPa/min; 6.H₂S≤20ppm; 8.Total Sulfur(In terms of sulfur)particle≤100mg/m³;9.Dust Granularity≤5μm; 10.Dust Content≤20mg/m³;11.Light oil content≤100mg/m³. Note: The standard reference condition for natural gas volume is 101.3kPa, 20°C.
Gas supply pressure(kPa)	Max. inlet 50 (Special pressure, contact with company)

Standard Features

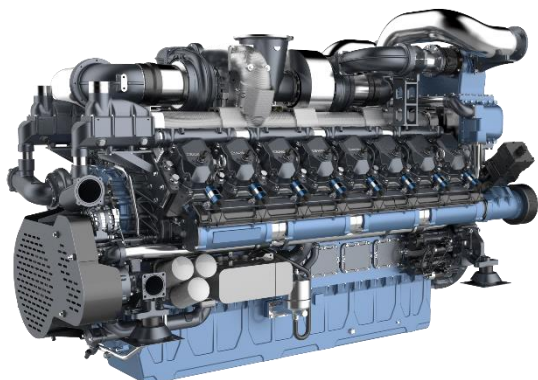
◆ ECU	◆ DC24 Electric starter	◆ WHC9510N
◆ Closed water-cooled	◆ Air filter	◆ Muffler
◆ Gas train	◆ Starting batteries	◆ Coupling
◆ Shock Absorber	◆ PMG	◆ Condensation heating
◆ Independence cabinet	◆ Generator bearing temperature monitor	◆ Generator winding temperature monitor
◆ Color	Weichai blue (B)/Container white(C)	
◆ Packaging	Plastic Film & Paulin Packaging	

Options

◇ Voltage380V/400V	◇ Voltage 415V/440V	◇ Voltage 6.3kV/10.5kV /13.8kV
◇ Single Voltage 220V/230V/240V	◇ Automatic Transfer Switch	◇ Multi motor radiator
◇ Engine Heater (water)		

Engine

- Weichai **16M33NG** series, 16 cylinder,V-type, 4 stroke, radiator cooled engine
- Well-designed air handling system with
 - Replaceable paper element air cleaner with restriction indicator
 - Air after cooling
 - Optimized turbocharger for increased altitude capabilities
 - High efficiency, large heat area intercooler, reduce the air temperature after the intercooler
- First fill of lube oil and coolant
- NG gas regulator
- NG gas filter
- Standard integral set-mounted radiator system, designed and tested for 40°C ambient temperature
- Electrical starter motor with soft start engagement feature
- Intelligent fuel injection technology to start fast guarantee
- Battery charger
- 24 V DC batteries



Alternator

- PMG, Screen protected, Revolving field, Self-excited alternator conforming to IEC 60034-1
- Best in class efficiency
- Steel casing
- Compact design with sealed bearings for longer life and lesser maintenance
- IP23 standard protection level
- Impregnation on all wound components for better mechanical strength



Control Module Specifications

The WHC9510N is an Auto Mains Failure Control Module

- Back-lit LCD display
- 3 Phase generator and 3 Phase Mains monitoring
- Monitoring speed, frequency, voltage, current, oil pressure, coolant temperature
- Display warning, shutdown and engine status information
- Hours counter provides accurate information for monitoring and maintenance



Engine specifications

Make	WEICHAI		
Mode	16M33D1280NG10		
Rated Speed r/min	1500		
Prime Power kW	1280		
Cooling	Close Liquid cooled		
Governor	ECU		
Aspiration	Turbocharged, After-cooled		
No. of cylinders	16, V Type.		
Bore (mm) x Stroke (mm)	150x185		
Displacement (L)	52.3		
Starting system	24VDC Electrical		
Total coolant capacity engine/tank (L)	130/NA		
Radiator design temperature (°C)	40		
Lubrication system capacity (L)	171		
Load rate (%)	100%	75%	50%
Fuel consumption (kg/h)	230.2	178.1	126.2
Engine high water heat quantity(kJ/s)	560.0	469.2	369.9
Engine high water flow(m³/h)	92.9	92.8	92.8
Engine high water inlet/outlet Tem. (°C)	85.1/91.3	86.3/91.5	86.8/90.9
Engine high water inlet/outlet Pre. (kPa)	-5.8/109	-5.9/110	-6.1/110
Exhaust gas heat quantity(kJ/s)	832.5	681.2	492.5
Exhaust gas flow (kg/h)	6264.0	4848.1	3339.2
Exhaust Temperature (°C)	459	475	502
Exhaust back pressure (mbar)	75		
Exhaust Temperature after recycle(°C)	Heating water120/0.5Mpa steam 190/1MPa steam 220		

Alternator specifications

Make	WEICHAI
Alternator Frame	WHA-1375-4/0.4
Exciter Type	PMG
Enclosure	IP23
Voltage regulation	≤±1%
Class of Insulation	H
Winding Pitch	2/3
Rotor	Double bearings

Remarks

Prime Rating (PRP) :
Prime power is defined as being the maximum power which a generating set is capable of delivering continuously while supplying a available electrical load when operated for an unlimited number of hours per year under the agreed operating conditions with the maintenance intervals and procedures being carried out as prescribed.

Continual Rating (COP):
These ratings are applicable for supplying continuous electrical power (at 100% load) in the event of a utility power failure. There is no limitation to the annual hours of operation under the agreed operating conditions with the maintenance intervals and procedures being carried out as prescribed.

Standard Reference Conditions

Standard operating environment: ambient temperature is -10℃~40℃, the altitude is less than 1000m,the relative humidity is less than 90%(25℃) and there is no dust\stand dust\salt fog\mold\condensation environment, etc.

Fuel consumption data at full load with gas fuel with specific gravity of 0.726 kg/N m³ and low calorific value of 47.9 MJ/kg.

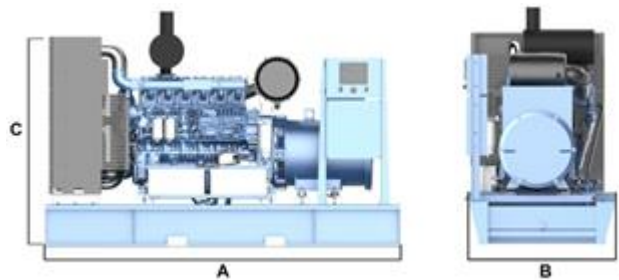
If the operating environment exceeds the above requirements, please contact with the factory.

Typical Genset Dimensions

Genset Model	Length A(mm)	Width B(mm)	Height C(mm)	Wet Weight (kg)	Note
WPG1375B7NG	6090	2300	2900	12300	
WPG1375C7NG	12192	2438	2896	19900	

Note: This size cannot be used for design, for design, please contact with company.

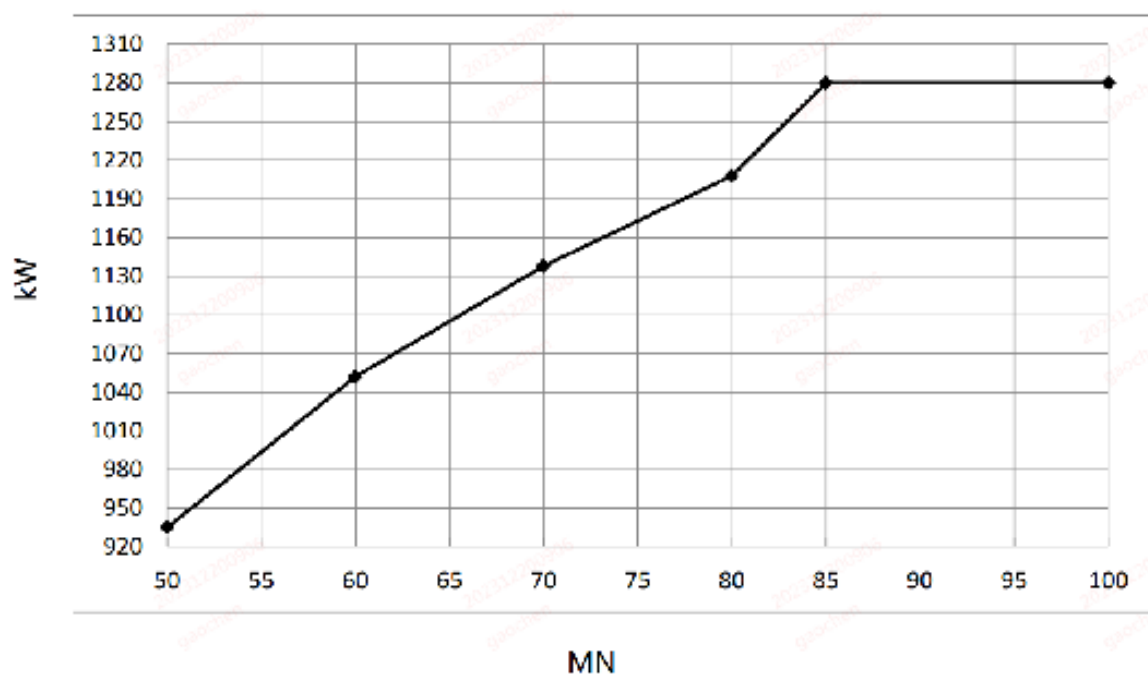
Open genset



Container genset



WEICHAI 潍柴	Model N° : 16M33D1280NG10	Date : 01/01/24
	Gas Engine MN Derating Curves	Page : 1 / 1



	Model N° : 16M33D1280NG10	Date : 01/01/24
	PowerKit Engine Derating Curves	Page : 1 / 1

